

i Front matter

Department of Electronic Systems

Examination paper for TFE4141 Design of Digital Systems 1

Date: December 15, 2025

Time (from-to): 15:00 – 19:00

Course contact: Per Gunnar Kjeldsberg

Present at the exam location: YES (approximately 16:30)

Permitted examination support material: C / Specified printed and hand-written support material is allowed. A specific basic calculator is allowed.

You are allowed to bring the course textbook (Dally, Harting and Aamodt, "Digital design using VHDL, a systems approach") as well as four (4) hand-written one-sided sheets of paper with your own notes.

OTHER INFORMATION

Get an overview of the question set before you start answering the questions.

Read the questions carefully and make your own assumptions. In your answers, explain clearly what assumptions you have made and how you have understood or limited the assignment

If there are direct errors or omissions in the assignment set and you cannot make your own assumptions, please refer to the information about complaints regarding formal errors on the NTNU website "Explanation of grades and appeals".

Paper drawings: For questions **13** and **14** it is possible to submit the entire/some parts of the answer as handwritten sheets. Other questions must be answered directly in Inspira. At the bottom of the question you will find a seven-digit code. Fill in this code in the top left corner of the sheets you wish to submit.

Please note that only blue or black ballpoint pens are allowed

We recommend that you do this during the exam. If you require access to the codes after the examination time ends, click "Show submission".

You are responsible for filling in the correct codes on the handwritten sheets. Therefore, read the cover sheet carefully. The Examination Office cannot guarantee that incorrectly completed sheets will be added to your assignment.

Weighting: The total number of achievable points is 100. The maximum number for each question is given on each question cover page.

Withdrawing from the exam: If you become ill or wish to submit a blank test/withdraw from the exam for another reason, go to the menu in the top right-hand corner and click "Submit blank". This cannot be undone, even if the test is still open.

Access to your answers: After the exam, you can find your answers in the archive in Inspira. Be aware that it may take a working day until any hand-written material is available in the archive.

1 **Simulation result (Correct answer 2 points, wrong answer -1 point, no answer 0 points)**

Given the following VHDL code. Which alternative timing diagram reflects a correct simulation result?

```

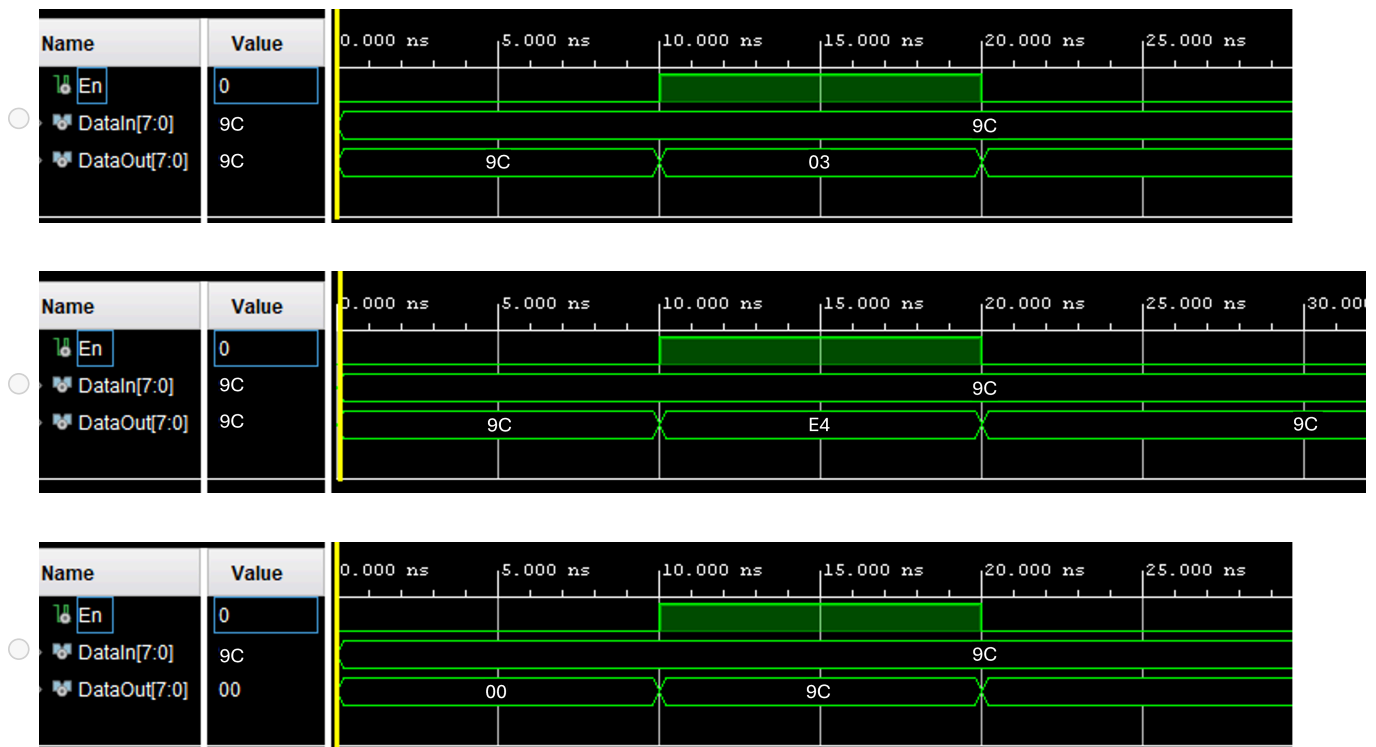
library IEEE;
use IEEE.NUMERIC_STD.ALL;

entity Task1 is
  Port ( En: in BIT;
        DataIn : in UNSIGNED (7 downto 0);
        DataOut : out UNSIGNED (7 downto 0));
end Task1;

architecture Behavioral of Task1 is
  constant Three: integer := 3;
begin
  DataOut <= (DataIn rol Three) when En = '1' else
    DataIn;
end Behavioral;

```

Select one alternative:



Maximum marks: 2

2 Resulting value (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Given the VHDL code below. Assume that we assign the values '1', '0', and 'Z' to insig1, insig2, and insig3, respectively. What is the resulting value on outsig?

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity Task2 is
    Port ( insig1, insig2, insig3: in STD_ULOGIC;
          outsig : out STD_LOGIC);
end Task2;

architecture Behavioral of Task2 is
begin
    outsig <= insig1;
    outsig <= insig2;
    outsig <= insig3;
end Behavioral;
```

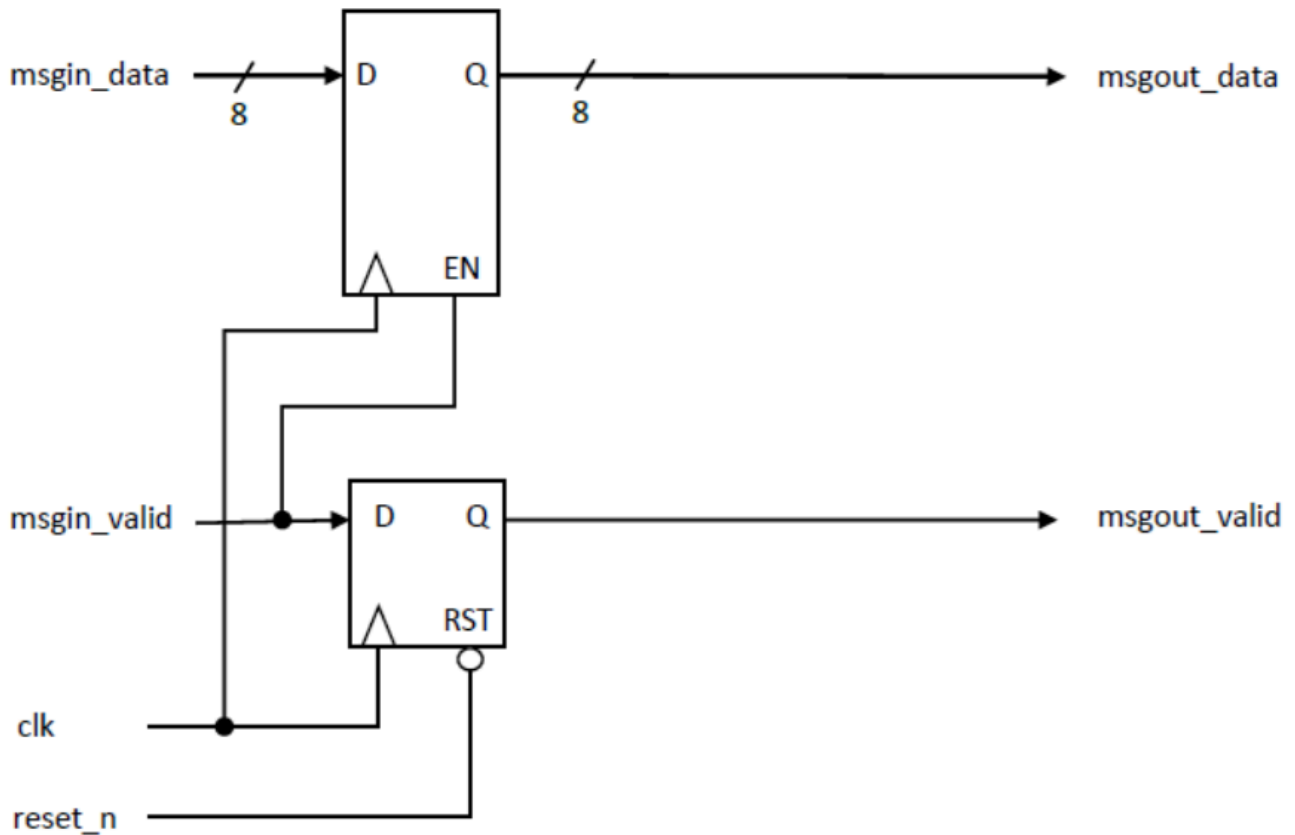
Select one alternative:

- ☐ 'Z'
- ☐ '1'
- ☐ 'X'

Maximum marks: 2

3 Matching representations (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

We have three representations of a digital circuit, a block diagram, a VHDL code, and a wave form. Which statement below reflects how the three representations match (i.e., they represent equivalent behaviours)?



```

library ieee;
use ieee.std_logic_1164.all;

entity Task3 is
  port( signal clk : in std_logic;
        signal reset_n : in std_logic;
        signal msgin_valid : in std_logic;
        signal msgin_data : in std_logic_vector(7 downto 0);
        signal msgout_valid : out std_logic;
        signal msgout_data : out std_logic_vector(7 downto 0) );
end Task3;

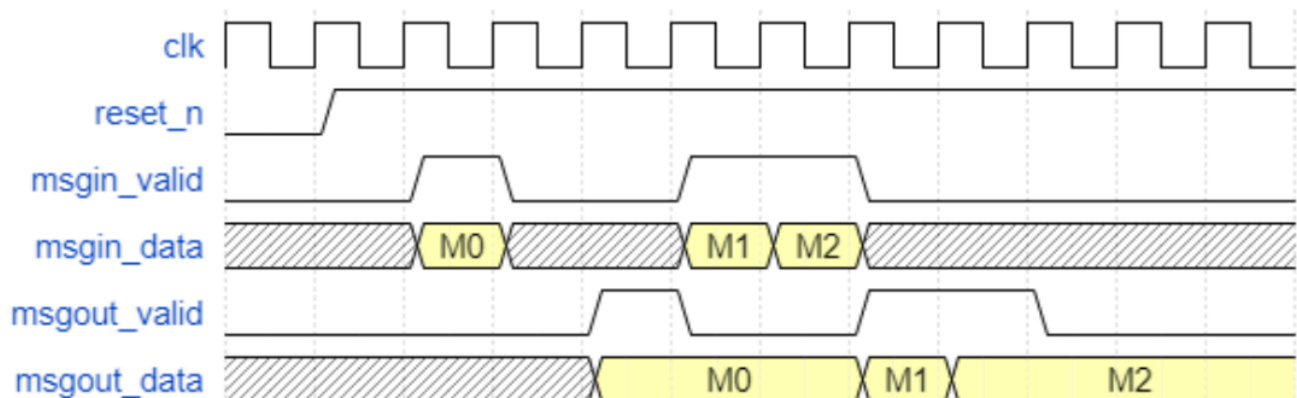
architecture Behavioral of Task3 is
begin

  process(clk) is
  begin
    if(clk'event and clk = '1') then
      if(msgin_valid = '1') then
        msgout_data <= msgin_data;
      end if;
    end if;
  end process;

  process(clk, reset_n) is
  begin
    if(reset_n = '0') then
      msgout_valid <= '0';
    elsif(clk'event and clk = '1') then
      msgout_valid <= msgin_valid;
    end if;
  end process;

end Behavioral;

```

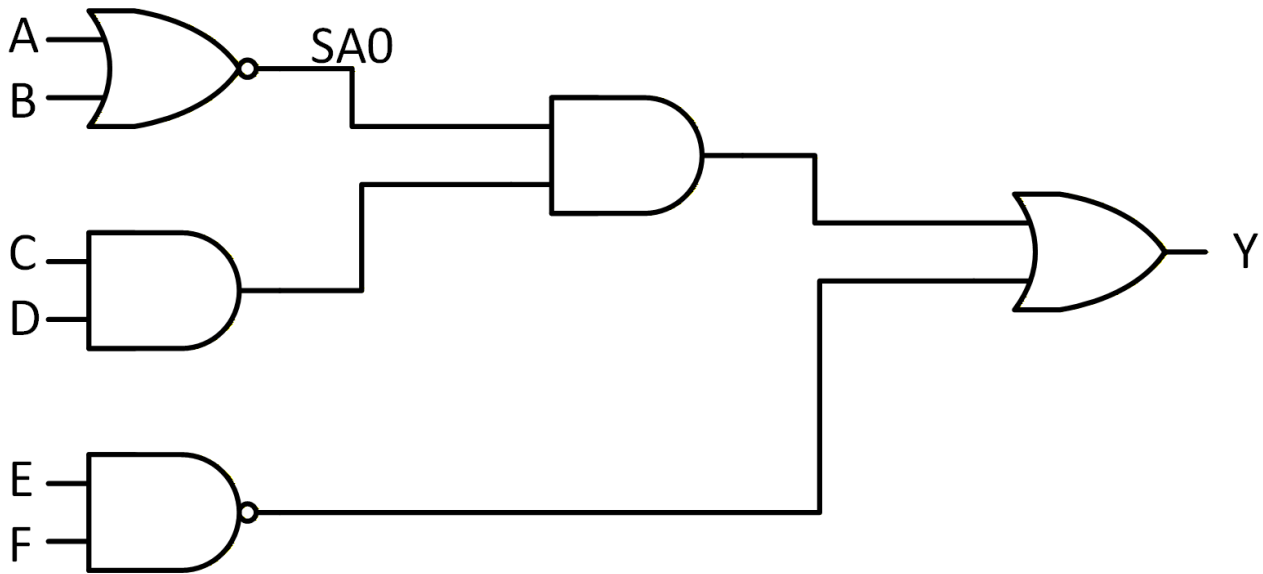


Select one alternative:

- ☐ The block diagram is matching both the VHDL code and the wave form
- ☐ The block diagram is only matching the VHDL code
- ☐ The block diagram is not matching any of the other representations

4 **Test pattern (Correct answer 2 points, wrong answer -1 point, no answer 0 points)**

Given the circuit below. Which statement is true related to a test for the output of the NOR-gate stuck at zero (SA0)?



Select one alternative:

- ☐ ABCDEF = 001111 is one of several possible test pattern
- ☐ There does not exist any test pattern for this stuck-at fault
- ☐ The only possible test pattern is ABCDEF = 001111

5 High level synthesis (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Given the following high-level code :

```
for ( i = 0; i <= 7; i++)
    Y = Y + A * indata[i] + B;
```

A and B are constant during one execution of the complete code but may change before a new execution. All input values (A, B, and each indata) are 8 bit vectors and can be accessed in parallel if needed.

Which statement is **correct** related to a digital implementation of the corresponding circuit?

Select one alternative:

- ☐ A fully combinatorial implementation requires 16 adders and 8 multipliers
- ☐ If we unroll the loop twice (partial unroll = 2) we will need (at least) four clock cycles to complete the calculation
- ☐ Any micro architecture requires at least one multiplier and two adders.

Maximum marks: 2

6 Testbench signals (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Given the following process written in a testbench. Which alternative timing diagram reflects a resulting wave form?

```
process
    constant period : time := 20 ns;
begin
    wait for (period * 0.7);
    stimuli <= '1';
    wait for (period * 0.3);
    stimuli <= '0';
end process;
```

Select one alternative:

- | Name | Value | 0.000 ns | 20.000 ns | 40.000 ns | 60.000 ns | 80.000 ns | 100.000 ns |
|-------------------------------|-------|----------|-----------|-----------|-----------|-----------|------------|
| ☐ stimuli | 0 | | | | | | |
| ☐ stimuli | 0 | | | | | | |
| ☐ stimuli | 0 | | | | | | |

Maximum marks: 2

7 **What component (Correct answer 2 points, wrong answer -1 point, no answer 0 points)**

What type of component does the VHDL code below implement?

```
use ieee.std_logic_1164.all;

entity Task7 is
    port( signal in1 : in std_logic;
          signal in2 : in std_logic;
          signal out1 : out std_logic);
end Task7;

architecture Behavioral of Task7 is
begin

    process(in1, in2) is
    begin
        if(in1 = '1') then
            out1 <= in2 after 2 ns;
        end if;
    end process;
end Behavioral;
```

Select one alternative:

- ☐ Multiplexer
- ☐ Flipflop
- ☐ Latch

Maximum marks: 2

8 **Correct statement (Correct answer 2 points, wrong answer -1 point, no answer 0 points)**

Which statement is correct?

Select one alternative:

- ☐ Due to its structured design, it is straight forward to perform a complete manufacturing test of a large multiplier.
- ☐ A scan chain can reduce the number of pins needed for manufacturing testing.
- ☐ When writing a testbench for design verification you need to make sure that it is synthesizable.

Maximum marks: 2

9 True and false statements (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

How many of the following two statements are **correct**?

- A generic constant can, for instance, be used to include different instants of the same module with different word widths in your design.
- A generate statement can, for instance, be used for simple structural repetition in your VHDL code.

Select one alternative:

- ☐ 1
- ☐ 2
- ☐ 0

Maximum marks: 2

10 True or false (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

How many of the following two statements are **correct**?

- Binding is a High Level Synthesis task where operations are assigned to operators. E.g., what multiplier should be used for a given multiplication in the code.
- Using as short abbreviations as possible for your signal names is in general a good idea in your VHDL code since the code then becomes more compact and easier to maintain.

Select one alternative:

- ☐ 2
- ☐ 1
- ☐ 0

Maximum marks: 2

Vdd [V]	2	1
Energy per cycle [nJ]	50	20
fmax [MHz]	50	10
cycle time [ns]	20	100

Fill in your answer here

Maximum marks: 12

12 Project phases (Maximum 8 points)

During digital design of a large system involving multiple teams of people, you typically have multiple phases in the project. Assume that your company starts with an *initial investigation*, followed by a *detailed investigation*, and finally *design & test*. Describe what activities typically takes place in each phase and, for *design & test*, what milestones you will expect to have.

Fill in your answer here

Format

B

I

U

$\times_2$

$\times^2$

$\frac{\square}{\square}$

































Σ



Words: 0

Maximum marks: 8

$$C = M^e \bmod n, M < n$$

Fill in your answer here

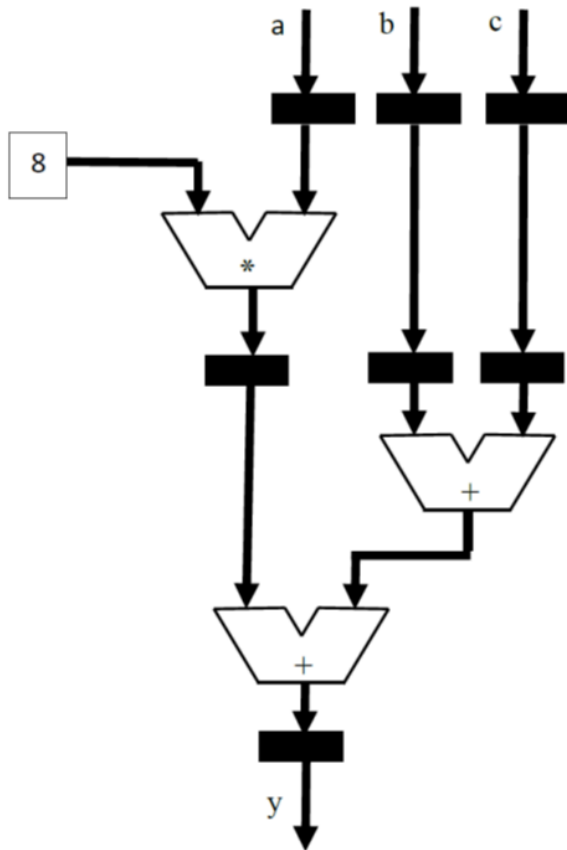
Words: 0

12/14

¹⁴ Design task (maximum 40 points)

Your boss has given you the block diagram below and asked you to be in charge of the design of the corresponding digital component. You are asked to improve both the area and the clock frequency of the design.

Inputs a, b and c are 16 bit wide signals of type unsigned. Black rectangles are registers of appropriate width. The square with the number "8" in it, indicates multiplication with the decimal number 8.



a) (max 10 points)

Explain how it is possible to modify the design above so that the clock frequency is doubled and the area is reduced. You should do so without adding any multiplexers. Draw a new block diagram reflecting the required changes.

b) (max 10 points)

Write synthesizable VHDL code that implements your design from a)

c) (max 10 points)

After having presented your design to your boss you are told that the inputs a, b, and c arrive sequentially on a common 16 bit wide bus. The system also includes a one bit signal START that indicates the arrival of input value a) and a one bit signal FINISHED which should indicate that the full calculation is finished.

You now need a state machine that controls the storing of data in registers and generates the FINISHED signal. Draw a state diagram and write synthesizable VHDL code for the state machine. Include a timing diagram that describes your assumptions regarding the relationship between the input and output signals.

d) (max 10 points)

Write a testbench that verifies the functionality of the complete design, applying at least two sets of inputs (i.e., the START signal should be applied twice) and compares the calculated result with the expected result.

For maximum score, perform verification of your design from Task 14 c). If you have not completed Task 14 c) you can use the original design from the figure above, but then the max number of points you can achieve is 7.

Fill in your answer here

[illegible]

Words: 0

Maximum marks: 40